

How raffle.fun Makes NFT Raffles Work Entirely Onchain

READER

Anyone who knows NFTs but not Solidity

PROTOCOL COMMIT

5772e54ba89c

NOT DEPLOYED

NOT INDEPENDENTLY AUDITED

Pre-release

A plain-language explanation of the complete sponsor and buyer experience. No Solidity required. Every factual claim here traces to [docs/facts/raffle-fun-facts.md](#), which cites the exact contract, function, and test behind it. Described commit: [5772e54ba89c06646815ed52a881cd8940f094ca](#).

1. The basic idea

You already know how a raffle works. Someone donates a prize, people buy numbered tickets, a number is drawn, and whoever holds that ticket takes the prize home.

Everything interesting happens in the gaps between those steps. Who holds the prize while tickets are selling? Who draws the number, and how do you know they did not pick a friend's ticket? What happens if barely anyone shows up, or if the organizer disappears?

In the physical world those gaps are filled by trust — in a charity, a licensing body, a friend. raffle.fun fills them with code: smart contracts on Base, an Ethereum layer-2 network, that hold the prize, sell the tickets, draw the number, pay out, and — importantly — refund everyone when something goes wrong. In one sentence: **raffle.fun locks one NFT into a contract, sells numbered tickets for a stablecoin, draws one ticket using an external randomness service, and lets whoever holds that ticket claim the prize.**

2. How the sponsor creates a raffle

The person offering the prize is the **sponsor**. They choose:

- **The prize** — one specific NFT they already own.
- **The ticket price** — a flat amount in the raffle's stablecoin, USDC in the intended deployment. (The contracts accept whichever single token is chosen when the factory is deployed; USDC is a deployment-time decision, not hard-coded into the raffle logic.)
- **A minimum ticket threshold** — how many tickets must sell for the NFT itself to be the prize. Below it, the raffle pays out in cash instead. Section 11 explains why.
- **A sale window** — the start can be up to seven days out, and the sale can run up to 30 days.
- **A recovery address** — where the NFT goes if the raffle does not award it. Defaults to the sponsor.

They approve the NFT and send one transaction. That is the entire setup: no application, no approval queue, no listing fee, no counterparty.

Note what the sponsor *cannot* do: cancel once a single ticket has sold; change the price, threshold, deadline, or prize; choose the winner; pause the draw; or withdraw early. Those are not policies someone enforces — they are functions that do not exist.

3. Why the NFT must be escrowed first

This is the most important structural decision in the protocol.

When the sponsor sends that creation transaction, five things happen inside it: a new contract is deployed for this specific raffle; it is recorded in an official registry; the NFT is transferred into it; the contract confirms it received the right NFT from the right person; and the registry confirms the NFT really arrived.

All five are one transaction, which in blockchain terms means **all of them succeed or none of them do**. If the NFT transfer fails, the deployment itself is undone along with the registry entry. There is no half-created raffle. So a raffle you can see is a raffle whose prize is already locked — you are never buying a ticket on a promise, unlike most offchain raffles where the prize stays with the organizer until the end.

The escrow is one-way. The raffle accepts an incoming NFT only if it is the exact configured token, sent by the sponsor, through the official factory, while still awaiting its prize. Anything else is rejected. Once the prize is in, no function exists — for anyone — that pulls it out except through the settlement paths below.

4. What buyers receive

A buyer approves the stablecoin and buys 1 to 100 tickets per transaction. The price shown is the price paid; there is no checkout fee. The buyer names the recipient, so you can buy tickets for someone else.

Tickets are numbered sequentially from 1 across all buyers: buy the first ten and you own 1 through 10, the next person starts at 11, and numbering never restarts or reuses a number. There is no cap on total tickets and none per buyer.

One detail matters more than it sounds: the contract checks its own balance before and after your payment and requires it to rise by *exactly* the amount owed. If the token delivered even one unit less — as some do by design — the purchase fails rather than issuing tickets the pot cannot back. This is why tokens with transfer taxes or rebasing supplies cannot be used at all.

5. Why tickets are ERC-721 tokens

Your tickets are not database rows on a website. They are NFTs, minted by the raffle contract itself, sitting in your own wallet, visible and movable like any other NFT. This has one enormous consequence:

The ticket is the claim. Whoever holds it at the moment of redemption is the person who can claim.

The contract never looks at who originally bought a ticket. It asks one question: are you, right now, the owner? If yes, you may burn it and take what it is owed. If no, you may not — and that includes anyone you have "approved" on a marketplace. An approval lets someone *move* your ticket; it does not let them *redeem* it. This is also why tickets can be traded at all: the claim travels with the token.

It is the main way a user can hurt themselves. Send a ticket to an address whose keys are lost, or to a contract that cannot call the redemption function, and the claim is gone forever. The protocol blocks the destinations it can recognize as dead ends — the raffle itself, the factory, the payment token, the randomness contract, the prize collection, and any other registered raffle — but it cannot screen every arbitrary contract without breaking ordinary NFT transferability, and deliberately does not try. No administrator can reverse the mistake, because there is no administrator.

6. When tickets may be transferred

Transferability changes with the raffle's state:

- **During the sale** — freely transferable. Trade, gift, list on a marketplace.
- **While randomness is pending** — **completely frozen**. Not one ticket can move.
- **After a result** — every ticket is transferable again *except the winning one*, which stays locked.
- **In refund mode** — everything is transferable again.

Section 9 explains why the freeze exists.

7. What happens when the sale ends

At the closing time, sales stop. Nothing else happens automatically — blockchains have no alarm clocks, and something only happens when someone sends a transaction.

So the next step is that **anyone** — sponsor, buyer, bot, bystander — pays the randomness service's current fee out of their own pocket and asks the raffle to draw. This is deliberately permissionless: a sponsor cannot stall a raffle they dislike by refusing to run the draw.

The catch is that the fee is real money in ETH and the protocol does not reimburse it. In practice a buyer with many tickets, or a sponsor expecting a payout, has every reason to pay it. But if nobody does within **three days** of the sale closing, the raffle gives up and refunds everyone. That is refund origin one.

If the raffle sold zero tickets, there is nothing to draw — see section 15.

8. How Pyth Entropy randomness works, at a high level

Picking a random number on a blockchain is famously hard, because everything on a blockchain is public and reproducible. If the raffle used something like a block hash, whoever produces blocks could quietly reroll until they liked the result.

raffle.fun uses **Pyth Entropy**, an external randomness service. The shape is a commit-and-reveal between the contract and an external provider: the request goes in, and later the provider's service calls back with a random value. The raffle computes the winning number as `random mod (tickets sold) + 1`.

Three honest caveats.

First, that mapping has a mathematically nonzero bias. When the ticket count does not divide evenly into the enormous range of random values, some numbers are very slightly more likely than others. For realistic ticket counts the difference is cryptographically negligible, but the protocol documents it rather than claiming perfect uniformity.

Second, and far more importantly, **the provider can see the result before publishing it**. Pyth documents this. A provider holding tickets could publish the answer if they won and stay silent if they did not — at which point the raffle times out and everyone, including them, is refunded. Heads they win, tails they get their money back. This is a documented, **unresolved** trust assumption, rated High severity by the project's own review; the release checklist requires pinning a reviewed provider or replacing the randomness design before any launch. Nobody should call this mechanism provably fair.

Third, the randomness step can simply fail to complete for boring reasons. Section 13 covers that.

9. Why transfers lock during the randomness period

Now the freeze makes sense.

Imagine transfers stayed open. The provider knows the winning number before anyone else. In the window between "the result exists" and "the result is public", someone with advance knowledge could buy the winning ticket from a seller who has no idea what they hold.

So the instant the draw is requested, **every ticket transfer is blocked**. Ownership is photographed at that moment. Whoever holds ticket 137 when the request goes in is the owner of ticket 137 if 137 is drawn — no marketplaces, no approvals, no exceptions.

Once the result publishes, ordinary tickets unfreeze. The *winning* ticket stays locked permanently, closing the mirror-image problem: a stale marketplace listing or forgotten approval could otherwise yank it out of the winner's wallet the instant the result became public. The winner cannot resell the winning ticket, but they can direct the prize to any safe address when they redeem it, which covers the legitimate version of that need.

Be clear about the limit: this stops someone *acquiring* the winning ticket after the draw. It does nothing about a provider who already owned tickets before it. That is the unresolved issue from section 8.

10. The NFT-success outcome

If the raffle sold at least its minimum threshold, the drawn ticket wins the NFT.

The winner calls one function, naming where the NFT should go. The contract burns their ticket, transfers the NFT, and then **verifies onchain that it actually arrived**. Only then does it calculate the protocol fee, credit the treasury, and credit the sponsor with the rest.

The sequencing is the point: the sponsor is paid not when the winner is *chosen* but when the winner is *delivered to*. Between resolution and redemption the entire pot sits in the raffle belonging to nobody in particular. If the transfer fails — collection paused, token burned, destination rejects it — the whole call reverts, the ticket is un-burned, nothing is paid, and the winner can retry elsewhere or later.

Worked example 1: the NFT branch

Sofia raffles a 1-of-1 NFT.

PARAMETER	VALUE
Ticket price	10.00 USDC
Minimum threshold	100 tickets
Tickets sold	250
Gross sales	2,500.00 USDC

The sale closes. A buyer named Maya pays the Entropy fee and requests the draw. Transfers freeze. The callback selects ticket **#137**, which Maya happens to hold.

Because $250 \geq 100$, this is the NFT branch. Maya redeems, naming her hardware wallet. Her ticket burns, the NFT arrives, and the contract verifies it. In that same instant:

RECIPIENT	AMOUNT
Protocol treasury (5% of 2,500.00)	125.00 USDC
Sponsor (Sofia)	2,375.00 USDC
Winner (Maya)	the NFT

Sofia and the treasury each now hold a claim they withdraw whenever they like. The other 249 tickets are worth nothing — souvenirs.

11. The cash-fallback outcome

The minimum threshold protects the sponsor from an otherwise common outcome: a valuable NFT going to someone who paid almost nothing because only a handful of tickets sold.

So if the sale finished **below** the threshold, the draw still happens and a ticket still wins — but it wins **money** instead, and the sponsor keeps the NFT plus the rest of the cash. Unlike the NFT branch, these amounts are recorded immediately when the result arrives, because no external NFT transfer has to succeed first.

Worked example 2: the cash branch

```
80 tickets sold
1.00 USDC each
minimum: 100 tickets ← missed
```

STEP	CALCULATION	AMOUNT
Gross sales	80×1.00	80.00 USDC
Protocol fee	5% of gross, rounded down	4.00 USDC
Distributable pot	$80.00 - 4.00$	76.00 USDC
Winning ticket	80% of the pot, rounded down	60.80 USDC
Sponsor	the remainder	15.20 USDC
Sponsor's recovery address		the NFT

The holder of the drawn ticket burns it and receives 60.80 USDC — a 60× return on a 1 USDC ticket, which is the point of the mechanism. Sofia gets her NFT back plus 15.20 USDC. The other 79 tickets get nothing.

On rounding. Both percentages round down and the remainder always goes to the sponsor, so nothing is lost or created. On an awkward gross of 999,999 raw token units, the fee is 49,999, the winner gets 760,000, and the sponsor 190,000 — exactly 999,999.

12. Refund origin one: nobody requested the draw

If the sale ends with tickets sold but no draw requested within **three days**, anyone can switch the raffle into refund mode, and the entire pot becomes refundable. Ticket holders burn their tickets — up to 100 per transaction — and receive **exactly the ticket price** for each. No protocol fee, no sponsor proceeds; the NFT returns to the sponsor's recovery address.

Refunds pay face value, not what you paid on a secondary market.

13. Refund origin two: the randomness never came back

If a draw request *was* accepted but no valid result arrives within **two days**, the same function opens the same full refund. Again: no fee, no sponsor proceeds, NFT returned.

One subtlety matters if you are watching closely: **a deadline does not change anything by itself**. At the two-day mark both a late-but-valid result and a refund finalization are legitimate; whichever lands first wins and the other becomes a harmless no-op. If nobody finalizes refunds, a valid result arriving a year later still settles the raffle normally. The project tests this deliberately so it cannot change by accident.

14. Refund origin three: the NFT could not be delivered

This one is the least obvious and, in some ways, the most important.

Suppose the raffle succeeded, the threshold was met, and a winner was drawn — but the NFT cannot be delivered. The collection got paused, the token was burned by its issuer, or the contract was upgraded into something broken.

An earlier design would have paid the sponsor at resolution and left buyers with a broken claim. raffle.fun instead keeps the whole pot escrowed until delivery is verified, and adds a deadline: if the NFT has not been claimed within **30 days** of the result, anyone can convert the raffle to full refunds. Every buyer gets the whole ticket price back; the sponsor and the protocol get nothing. A raffle where the prize cannot be delivered is a raffle that did not happen, so nobody should profit from it.

This also fires when a winner never shows up — thirty days of inaction turns an NFT win into a full refund for everyone, including the absent winner. As with the other deadline, a winner who resolves their problem on day 31 can still redeem, provided nobody finalized refunds first.

Worked example 3: the refund branch

Take exactly the setup from example 1 — 250 tickets at 10.00 USDC, 2,500.00 USDC in the pot — but this time nobody pays the Entropy fee in the three days after the sale closes.

RECIPIENT	AMOUNT
Protocol treasury	0.00 USDC
Sponsor	0.00 USDC
Sponsor's recovery address	the NFT
Every ticket holder	exactly 10.00 USDC per ticket
Total refunded	2,500.00 USDC

Maya, holding 40 tickets, redeems once with all 40 IDs and receives 400.00 USDC. If she held 250 she would need three transactions, because the batch limit is 100. Two practical notes: the batch is all-or-nothing, so a single duplicate or wrong ticket ID fails the whole transaction and consumes nothing; and nobody's refund depends on anyone else claiming theirs.

15. When nobody buys anything

If a raffle sells zero tickets, it never needs a draw. The sponsor can close it early at any time; once the sale window has ended, anyone can close it. The NFT goes back to the recovery address. That early-close privilege disappears the moment a single ticket sells.

16. Sponsor and treasury claims

The protocol never pushes money at anyone. It records what it owes and each party withdraws their own — the standard "pull payments" pattern, which stops one recipient's broken wallet from blocking everyone else's payout. A helper lets a third party pay the gas to settle someone else's claim, but it can only send those funds to the account owed them.

The NFT recovery path is separate and tighter. Only the recovery address named at creation can pull the NFT out, and only in the three states where the raffle did not award it: cash outcome, refund, or empty closure. It is deliberately unavailable in the NFT-win state, because there the prize belongs to the winner. That address is fixed forever at creation — a sponsor who names the wrong one has no remedy.

17. A complete numeric example, end to end

One raffle, every stage, with the money tracked the whole way.

Setup. Sofia raffles *Vault Key #7*: 25.00 USDC per ticket, minimum 200, seven-day sale, recovery address her cold wallet. One transaction deploys the raffle, registers it, and moves the NFT into escrow. Sofia can now touch neither the NFT nor any parameter.

Sale. Over seven days, 640 tickets sell to 47 addresses. Gross sales are $640 \times 25.00 = 16,000.00$ USDC, all held by the raffle. During the sale its books read:

LEDGER LINE	AMOUNT
Unsettled pot	16,000.00
Refunds owed	0.00
Winner cash owed	0.00

LEDGER LINE	AMOUNT
Sponsor + treasury claims	0.00
Total accounted	16,000.00

The contract's balance always equals or exceeds that total; anything extra, such as a direct donation, is surplus and never becomes anyone's claim.

Draw. The sale closes. Eleven hours later a ticket holder named Leo pays the Entropy fee in ETH and requests the draw; he overpays slightly and the excess returns in the same transaction. All 640 tickets freeze.

Result. The callback selects ticket **#411**, held by Noor. Because $640 \geq 200$, this is the NFT branch. Nothing is paid out yet — every ledger line above is unchanged, with the full 16,000.00 still in the unsettled pot.

Redemption. Noor redeems, naming her cold wallet. Her ticket burns, *Vault Key #7* arrives, the contract verifies ownership, and only then does settlement happen:

RECIPIENT	CALCULATION	AMOUNT
Protocol treasury	$\text{floor}(16,000.00 \times 5\%)$	800.00 USDC
Sponsor (Sofia)	$16,000.00 - 800.00$	15,200.00 USDC
Winner (Noor)		<i>Vault Key #7</i>

The books now read:

LEDGER LINE	AMOUNT
Unsettled pot	0.00
Refunds owed	0.00
Winner cash owed	0.00
Sponsor + treasury claims	16,000.00
Total accounted	16,000.00

Withdrawal. Sofia withdraws her 15,200.00 and the treasury its 800.00, independently and in either order. Every line goes to zero and the raffle is finished. The 639 losing tickets stay in their owners' wallets as souvenirs.

18. What "immutable" means for an individual raffle

This word gets used loosely, so here is exactly what it means.

Each raffle is its own freshly deployed contract — not a proxy, not a clone pointing at shared logic, but a standalone contract with its entire configuration frozen at deployment. There is no upgrade function and no shared implementation that could be swapped underneath it.

Concretely, once a raffle exists, **nobody** — sponsor, protocol team, factory owner, treasury — can change its prize, price, threshold, deadlines, fee rate, recovery address, payment token, or randomness

source; pause, cancel, or reverse it; choose or override a winner; or withdraw assets outside the settlement paths in this article.

That guarantee cuts both ways. The rules you read when you bought are the rules that apply. But if something goes wrong in a way the contract does not anticipate, **there is no help available** — no emergency pause, no admin refund, no support team who can move your ticket back. The absence of a rescue function is what makes the guarantee credible, and it is also what makes user error final.

19. What the factory owner can and cannot do

There is one administrative role in the protocol: the owner of the **factory**, the contract that creates raffles. That owner can do exactly two things, both applying only to raffles created *in the future*:

1. **Change which treasury receives the protocol fee.** Every existing raffle captured its treasury address permanently at creation, so this affects only new ones.
2. **Pause creation of new raffles.** This touches no existing raffle's sale, draw, refund, or claim.

Ownership transfer is two-step — a proposed owner must actively accept — so a typo cannot strand the role. Ownership also **cannot be renounced**: an earlier version allowed it, and the project's own review found that renouncing while creation was paused would have permanently bricked raffle creation. It now always reverts.

What the owner cannot do is more interesting: nothing at all to an existing raffle. Not pause it, seize its assets, change its fee, pick its winner, or redirect its payouts. If a serious bug were found, the realistic response would be to pause new creation, warn users, take down the website, and deploy a new factory. Raffles already running would run to their own conclusions, bug and all.

20. Risks and limitations

Read this before putting money at risk.

The randomness provider is trusted. Per section 8: the provider can know the result before publishing it, and if it holds tickets it could publish only when it wins. The project's own model puts the gain at up to a quarter of the pot, when the provider holds half the tickets. Unresolved, and rated High severity.

The stablecoin issuer is trusted. Circle can freeze, blacklist, pause, or upgrade USDC. If your address were blacklisted, no contract could force a payment to you. The contracts preserve your claim across a failed transfer so a later retry works — but they cannot make a frozen token move.

The chain is trusted. Base's sequencer orders transactions. It could delay a draw request past its three-day window or a callback past its two-day window, converting a raffle into refunds, and could in principle censor a user. A halted or reorganized chain has no application-level escape hatch.

The prize NFT is trusted. The protocol checks that a prize contract claims to be an ERC-721 and verifies delivery — but performs those checks *through the same contract that could be lying*. A malicious or upgradeable collection can misreport ownership, and can simply be worthless. **No smart contract can tell you whether an NFT is worth buying tickets for.** That judgment is yours.

Ticket destinations are your responsibility. A ticket sent to a lost key or an incapable contract is gone, with no recovery.

Nothing is private. Sponsor identity, every purchase and transfer, the winner, and every amount are permanently public. Websites and RPC providers may also see your IP address and wallet metadata.

The legal position is unresolved. Raffles, lotteries, sweepstakes, and prize promotions are regulated very differently across jurisdictions, and no jurisdiction-specific legal review has been performed. Whether participating in or sponsoring one is lawful where you live is a question this article cannot answer.

Losing tickets keep existing. Only the winning ticket is burned. Everyone else's tickets stay in their wallets and stay tradable after the raffle is over, even though they are now worth nothing. If you buy a ticket on a secondary market, check that the raffle is still running — a settled loser looks exactly like a live ticket.

Some things are unsupported outright. Tokens with transfer taxes or rebasing supplies are rejected; stablecoins donated directly to a raffle are unrecoverable; an unrelated NFT force-sent into a raffle has no rescue path. Using another raffle's ticket as a prize is a known bad idea: if that inner raffle locks the ticket, the prize can be stranded for good, though buyers still get refunded.

21. Current status

This is the part most likely to be misread, so it is stated plainly.

raffle.fun is not deployed. There is no live raffle.fun on Base or any other public network. The repository contains no deployed address, and the web application disables all write actions when no deployment record is present. Every test against real chain state has run on local forks; no transaction has been broadcast.

The contracts have not been independently audited. The project's own security policy says so in as many words: it "remains independently unaudited". The most recent review was self-administered, and its own scope note calls it "not an independent audit or a production authorization".

The project's own release checklist says, verbatim, "Current status: not release-ready." Outstanding blockers include an external audit, a monitored testnet period, selection of the multisig wallets that would own the factory and receive fees, production monitoring and incident-response runbooks, a staffed disclosure process, and jurisdiction-specific legal review.

What *has* happened is a substantial internal testing campaign — fuzzing, stateful and strict invariants, an independent differential model, two separate independent fuzzing tools, symbolic execution, mutation testing, static analysis, compiler differentials, and pinned fork validation against live Base state, measured in millions of generated call sequences, with roughly 99.7% line and 100% function coverage. The maintainers are explicit that this is "evidence, not a mathematical proof", and not a substitute for an independent audit — an unusual amount of self-imposed rigor for this stage, but still not an audit, and this article will not pretend otherwise.